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Method and system to obtain cytology image in cytopathology

Abstract

Example methods and systems to obtain images associated with target cells distributed in a cytology specimen have been disclosed. One example method includes obtaining a first image associated with a first region of the cytology specimen through a first object lens, determining a first subregion of the first region, obtaining a second set of one or more images associated with the first subregion through a second object lens, identifying a third image from the first image, and in response to a ratio of a second number of the target cells in the second set of one or more images to the first number of the target cells in the third image being in a predetermined range, obtaining images associated with the target cells based on the second set of images.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of and claims priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/602,278, filed Oct. 7, 2021, which is a 371 application of International Application PCT/US2021/03441, filed May 27, 2021, which claims the benefits of U.S. Provisional Application No. 63/093,287 filed Oct. 18, 2020 and U.S. Provisional Application No. 63/093,444 filed Oct. 19, 2020. The U.S. Patent Application, U.S. Provisional Applications, and International Application are incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) Embodiments of the present invention generally relate to methods and systems to obtain cytology images in cytopathology.

Description of the Related Art

(2) Unless otherwise indicated herein, the approaches described in this section are not prior art to the claims in this application and are not admitted to be prior art by inclusion in this section.

(3) Cytopathology is a branch of pathology that studies and diagnoses diseases on a cellular level and generally involves obtaining cytology images on the cellular level. Obtaining cytology images of a cytology specimen includes an image digitalization of a glass slide on which the cytology specimen is distributed. The image digitalization may generally include scanning the glass slide for the cytology specimen to generate an image digital slide for the glass slide. The scanning may be performed by a whole-slide imaging scanner. The image digital slide can be viewed on a display (e.g., computer monitor) instead of a microscope.

(4) Performing image digitalization of a glass slide efficiently and precisely can be challenging. One reason is that the cytology specimen on the glass slide may contain single cells and cell groups distributed in a three-dimensional manner. The three-dimensional distributions of the cells and cells groups in the cytology specimen cause focusing difficulties. One conventional way to address such focusing difficulties is to obtain focused images of the entire glass slide, which can be very time consuming. Another conventional way to address the focusing difficulties is to manually mark regions of the glass slide to be focused on and obtain focused images of the marked regions. However, the manual approach may mark incorrect regions and as a result, the obtained focused images may not include many target cells in the cytology specimen.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an example figure showing a cytology specimen distributed in a three-dimensional

manner on glass slide;

(2) FIG. 2 illustrates how an example multiple object lens module obtains images associated with a layer of a cytology specimen by a first object lens;

(3) FIG. 3A is an example figure illustrating a part of cytology specimen;

(4) FIG. 3B illustrates an image associated with a part of cytology specimen obtained by first object lens;

(5) FIG. 4A illustrates how an example multiple object lens module obtains images associated with a secondary subpart of a cytology specimen by a second object lens;

(6) FIG. 4B illustrates an image associated with secondary subpart obtained by second object lens;

(7) FIG. 4C illustrates how an example multiple object lens module obtains images associated with a plurality of secondary subparts of a cytology specimen by a second object lens;

(8) FIG. 4D illustrates an image associated with secondary subpart obtained by second object lens;

(9) FIG. 5 illustrates an example system to obtain images associated with a cytology specimen; and

(10) FIG. 6 is a flow diagram illustrating an example process to obtain images associated with target cells distributed in a cytology specimen, all arranged in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

(11) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the Figures, can be arranged, substituted, combined, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

(12) FIG. 1 is an example figure showing cytology specimen **110** distributed in a three-dimensional manner on glass slide **120**, arranged in accordance with some embodiments of the present disclosure. Cytology specimen **110** may include multiple cells and impurities. For example, cytology specimen **110** may include dust or mark **131**, target cells **141** (e.g., the Epithelial cells at malignant risk), non-target cells **151**, **153**, **155**, **157** and **159** (e.g., red blood cells, normal Epithelial cells and other cells). Dust or mark **131**, target cells **141** and non-target cells **151**, **153**, **155**, **157** and **159** are distributed in a three-dimensional manner (i.e., at different depths within the volume of cytology specimen **110**) on glass slide **120**.

(13) FIG. 2 illustrates how an example multiple object lens module **200** obtains images associated with a layer of cytology specimen **210** by first object lens **220**, arranged in accordance with some embodiments of the present disclosure. In conjunction with FIG. 1, in FIG. 2, cytology specimen **210** may correspond to cytology specimen **110**. In FIG. 2, multiple object lens module **200** includes first object lens **220** configured to obtain images associated with cytology specimen **210**. More specifically, first object lens **220** is configured to obtain images associated with one or more layers of cytology specimen **210**. In some embodiments, each layer may correspond to a depth of field associated with a field of view of first object lens **220**.

(14) In some embodiments, first object lens **220** has a first field of view, a first depth of field and a first magnification. The first field of view may correspond to any of region A, B, C, D, E, F, G, H, I and J of cytology specimen **210**.

(15) In some embodiments, in the first field of view, a distance between a sharp and focused object nearest to first object lens **220** and a sharp and focused object furthest to first object lens **220** refers to the first depth of field. Some example distances may be distance **1**, distance **2** and distance **3** illustrated in FIG. 2. The first depth of field may also correspond to a layer of cytology specimen **210**. Therefore, distance **1** may correspond to the first layer of cytology specimen **210**; distance **2** may correspond to the second layer of cytology specimen **210**; and distance **3** may correspond to

the third layer of cytology specimen **210**.

(16) The first layer may be a layer furthest away from a glass slide (e.g., glass slide **120**) and the third layer may be a layer adjacent to the glass slide (e.g., glass slide **120**). Noting that although FIG. 2 illustrates 3 layers, cytology specimen **210** may include more or less layers according to the first depth of field of first object lens **220**. In these embodiments, a region and a distance may define a part of cytology specimen **210**. For example, **J3** corresponds to a front bottom right part of cytology specimen **210**. Therefore, in one embodiment illustrated in FIG. 2, cytology specimen **210** may be defined by parts **A1, A2, A3, B1, B2, B3, C1, C2, C3, D1, D2, D3, E1, E2, E3, F1, F2, F3, G1, G2, G3, H1, H2, H3, I1, I2, I3, J1, J2** and **J3**.

(17) In some embodiments, cytology specimen **210** may include, but not limited to, dust **211**, mark **212**, first type of cells **213** (e.g., the Epithelial cell at malignant risk), second type of cells **214** (e.g., red blood cell), third types of cells **215** (e.g., normal Epithelial cell) and fourth types of cells **216** (e.g., other cells). Dust **211**, first type of cells **213**, second type of cells **214**, third types of cells **215** and fourth types of cells **216** may be distributed in cytology specimen **210** based on their specific gravities. Mark **212** may be manually marked by a user on a top surface of cytology specimen **210**. Dust **211** and cells **213, 214, 215** and **216** have significant differences in their specific densities and may be distributed in different layers of cytology specimen **210**. For example, dust **211** has a smaller specific density than cells **213, 214, 215** and **216** and may be distributed in the first layer of cytology specimen **210**. Cells **213, 214, 215** and **216** have specific densities that are larger than dust **211** and may be distributed in the second layer of cytology specimen **210**.

(18) In some embodiments, cells **213, 214, 215** and **216** in cytology specimen **210** usually have sizes in a range of around 5 micrometres to around 20 micrometres. It should be noted that a selection of first object lens **220** may be based on the first depth of field of first object lens **220**. The first depth of field of first object lens **220** may be much larger than the sizes of cells **213, 214, 215** and **216** in cytology specimen **210** so that the images of cells **213, 214, 215** and **216** are likely to be obtained in the same first depth of field (e.g., distance 1, distance 2 or distance 3).

(19) In some embodiments, prior to obtaining images of an entire layer of cytology specimen **210** through first object lens **220**, first object lens **220** is configured to obtain images of each layer (e.g., first layer, second layer and third layer) of several regions randomly selected from all regions **A, B, C, D, E, F, G, H, I** and **J**. For example, first object lens **220** may be configured to obtain images of parts **H1, H2** and **H3** in the randomly selected region **H**. Here, region **H** corresponds to the first field of view of first object lens **220**, and distance 1, 2 or 3 corresponds to the first depth of field of object lens **200**. Several regions other than region **H** may be also randomly selected, for example, regions **F, I** and **J**. Accordingly, first object lens **220** may also obtain images of parts **F1, F2, F3, I1, I2, I3, J1, J2** and **J3** of cytology specimen **210**.

(20) In some embodiments, images of parts **F1, F2, F3, H1, H2, H3, I1, I2, I3, J1, J2** and **J3** are processed by a processor with an artificial intelligent engine to identify whether the images of parts **F1, F2, F3, H1, H2, H3, I1, I2, I3, J1, J2** and **J3** include images of cells **213, 214, 215** and **216**. The artificial intelligent engine may include machine learning capabilities. The artificial intelligent engine may be trained based on sample images of known cells corresponding to cells **213, 214, 215** and **216** having various contrasts, lightness, shapes and other image characteristics. In some embodiments, for illustration purposes only, the artificial intelligent engine may identify images of cells **213, 214, 215** and **216** that are mostly included in images of parts **F2, H2, I2** and **J2**, instead of being included in images of parts **F1, F3, H1, H3, I1, I3, J1** and **J3**. Accordingly, the artificial intelligent engine may conclude that cells **213, 214, 215** and **216** are distributed in the second layer of cytology specimen **210**. After reaching such a conclusion, first object lens **220** is configured to obtain images of the entire second layer of cytology specimen **210** (i.e., images of parts **A2, B2, C2, D2, E2, F2, G2, H2, I2** and **J2**).

(21) In conjunction with FIG. 2, FIG. 3A is an example figure illustrating a part **310** of cytology specimen **210**, and FIG. 3B illustrates an image **330** associated with part **310** obtained by first

object lens **220**, arranged in accordance with some embodiments of the present disclosure. In some embodiments, part **310** may correspond to part H2 of cytology specimen **210**. Cells **321**, **322**, **323**, **324**, **325** and **326** may be distributed in part **310**. In some embodiments, cells **324** may be the Epithelial cells at malignant risk and cells **321**, **322**, **323**, **325** and **326** may include normal Epithelial cells, red blood cells and other cells.

(22) In some embodiments, image **330** may be a top view of part **310** because first object lens **220** is disposed at the top of part **310**. Image **330** in FIG. 3B has an image region H' corresponding to region H associated with part **310** in FIG. 3A. The correspondence between region H and image region H' may include one or more factors associated with enlargement, shrinkage, rotation or twisted.

(23) In some embodiments, the artificial intelligent engine set forth above is used to process image **330** based on contrasts, lightness, shapes and other image characteristics associated with cells **321** to **326** to identify images **331**, **333** and **335** of image **330** which include images of cells **321** to **326**. For example, images associated with cells **321** and **322** are identified to be included in image **331**, images associated with cells **323**, **324** and **326** are identified to be included in image **333** and images associated with cells **321**, **322** and **325** are identified to be included in image **335**. Any of images **331**, **333** and **335** has a spatial relationship in image region H' of image **330**. Based on the spatial relationship between images **331**, **333**, **335** and image region H', and the correspondence between image region H' and region H, referring back to FIG. 3A, subregions **341**, **343** and **345** in region H may be identified. In some embodiments, subregions associated with cells respectively distributed in any of part A2, B2, C2, D2, E2, F2, G2, **12** and J2 may be identified in a similar manner.

(24) In some embodiments, subparts **351**, **353** and **355** of part **310** are defined by distance **2** and subregions **341**, **343** and **345**, respectively. Subparts **351**, **353** and **355** are further examined with a second object lens to obtain more detailed images associated with cells **321** to **326**. In some embodiments, as set forth above, cells **324** may be the Epithelial cells at malignant risk (i.e., target cells) and cells **321**, **322**, **323**, **325** and **326** may include normal Epithelial cells, red blood cells and other cells (i.e., non-target cells), subpart associated with the target cells (i.e., subpart **353**) may be examined in more details than subparts associated with the non-target cells (i.e., subparts **351** and **355**). For example, the second object lens is configured to obtain images associated with subpart **353** in more depths of field than subparts **351** and **355**. For example, the second object lens may obtain images associated with subpart **353** in different two depths of field while obtaining an image associated with subpart **351** in one single depth of field and an image associated with subpart **355** in one single depth of field.

(25) In some embodiments, as set forth above, subparts of parts A2, B2, C2, D2, E2, F2, G2, **12** and J2 may also be further examined with the second object lens to obtain more detailed images associated with cells **321** to **326**.

(26) FIG. 4A illustrates how an example multiple object lens module obtains images associated with a secondary subpart of a cytology specimen by a second object lens, arranged in accordance with some embodiments of the present disclosure. In FIG. 4A, second object lens **410** of the multiple object lens module is configured to obtain one or more images associated with subpart **420**. In conjunction with FIG. 3A, subpart **420** may correspond to subpart **353** defined by distance **2** and subregion **343**.

(27) In some embodiments, second object lens **410** has a second field of view, a second depth of field and a second magnification. In some embodiments, in conjunction with FIG. 3A, the second field of view may correspond to subregion **343**. In some other embodiments, in conjunction with FIG. 3A, the second field of view may correspond to only a part of subregion **343**. For illustration only, FIG. 4A illustrates that the second field of view corresponds to subregion **343**.

(28) In some embodiments, the second magnification is higher than the first magnification. For illustration only, the first magnification and the second magnification may be, but not limited to, 4×

and 20×, respectively.

(29) In some embodiments, the second depth of field is less than the first depth of field. For illustration only, the first depth of field and the second depth of field may be, but not limited to, around 50 micrometres and around 1 micrometre, respectively. In some embodiments, in the second field of view, a distance between a sharp and focused object nearest to second object lens **410** and a sharp and focused object furthest to second object lens **410** refers to the second depth of field. Some example distance may be distance **21**, distance **22**, distance **23** or distance **24** of distance **2** illustrated in FIG. **4A**. Distance **2** illustrated in FIG. **4A** corresponds to same distance **2** illustrated in FIG. **2** and FIG. **3A**.

(30) In some embodiments, the second depth of field may also correspond to a sublayer of a layer (e.g., second layer defined by distance **2** illustrated in FIG. **2**) of cytology specimen **210**. Therefore, distance **21** may correspond to a first sublayer of the second layer of cytology specimen **210**, distance **22** may correspond to a second sublayer of the second layer cytology specimen **210**, distance **23** may correspond to a third sublayer of the second layer of cytology specimen **210** and distance **24** may correspond to a fourth sublayer of the second layer of cytology specimen **210**. Noting the number of sublayers of **4** is only for illustration and may include more or less sublayers according to the second depth of field of second object lens **410**.

(31) In some embodiments, second object lens **410** is configured to obtain images associated with subpart **420**. More specifically, second object lens **410** is configured to obtain images of secondary subparts **421**, **422**, **423** and/or **424** of subpart **420**. In some embodiments, second object lens **410** is configured to focus on secondary subpart **421**, subpart **422**, subpart **423** and/or subpart **424** with the second depth of field.

(32) In some embodiments, the focus may be based on a determination of an artificial intelligent engine. The artificial intelligent engine may include machine learning capabilities. For example, the artificial intelligent engine may be trained based on sample images of target cells (e.g., the Epithelial cells at malignant risk) having various contrasts, lightness, shapes and other image characteristics. Accordingly, the artificial intelligent engine may identify images of the target cells and determine to focus on these identified images. For example, among images of secondary subparts **421**, **422**, **423** and **424** obtained by the second object lens **410**, the artificial intelligent engine may identify images of secondary subparts **422** and **423** include images of the target cells. Based on the determination made by the artificial intelligent engine, the second object lens **410** is then driven to focus on secondary subparts **422** and **423** and save images of secondary subparts **422** and **423**.

(33) In alternative embodiments, the focus may be based on image characteristics of target cells (e.g., the Epithelial cells at malignant risk). Such image characteristics may include, but not limited to, color information associated with the target cells. For example, the nuclei of an Epithelial cell may be stained to blue by hematoxylin while red blood cells maintain red because red blood cells have no nuclei to be stained. Accordingly, focusing on blue parts, instead of red parts, in the image have a higher chance to obtain images of the target cells.

(34) In some embodiments, images of target cells may include a first specific range in the RGB (Red, Green, Blue) domain or in the HSV (Hue, Saturation, Value) domain. For example, the first specific range may include a range of R, G and B values, respectively. Alternatively, the first specific range may include a range of Hue value. On the other hand, images of non-target cells (e.g., normal Epithelial cells, red blood cells or other cells) may include a second specific range in the RGB (Red, Green, Blue) domain or in the HSV (Hue, Saturation, Value) domain, which is different from the first specific range. For example, the second specific range may include another range of R, G and B values. Alternatively, the second specific range may include another range of Hue value. For example, among images of secondary subparts **421**, **422**, **423** and **424** obtained by the second object lens **410**, images of secondary subparts **422** and **423** may include the first specific range. Accordingly, the second object lens **410** is then configured to focus on secondary subparts

422 and **423** and save images of secondary subparts **422** and **423**.

(35) FIG. **4B** illustrates an image **430** associated with secondary subpart **422** obtained by second object lens **410**, arranged in accordance with some embodiments of the present disclosure. Image **430** includes images of target cell **441** and non-target cells **443** and **445**. Image of target cell **441** includes the first specific range in the RGB domain or in the HSV domain and images of non-target cells **443** and **445** include the second specific range in RGB domain or in HSV domain. The first specific range in the RGB domain or in the HSV domain associated with image of target cell **441** causes second object lens **410** to focus on secondary subpart **422**.

(36) In some embodiments, in conjunction with FIGS. **2** and **3A**, image **430** obtained by second object lens **410** is compared to image **333** obtained by first object lens **220** to determine whether a ratio of a number of target cells in image **430** to a number of target cells in image **333** is in a predetermined range. In conjunction with FIG. **3A**, in response to the ratio being in the predetermined range, it may indicate that image **430** obtained by second object lens **410** has captured most target cells in subpart **353**.

(37) FIG. **4B** illustrates an image **430** associated with secondary subpart **422** obtained by second object lens **410**, arranged in accordance with some embodiments of the present disclosure. Image **430** includes images of target cell **441** and non-target cells **443** and **445**. Image of target cell **441** includes the first specific range in the RGB domain or in the HSV domain and images of non-target cells **443** and **445** include the second specific range in RGB domain or in HSV domain. The first specific range in the RGB domain or in the HSV domain associated with image of target cell **441** causes second object lens **410** to focus on secondary subpart **422**.

(38) In some embodiments, in conjunction with FIGS. **2** and **3A**, image **430** obtained by second object lens **410** is compared to image **333** obtained by first object lens **220** to determine whether a ratio of a number of target cells in image **430** to a number of target cells in image **333** is in a predetermined range. In conjunction with FIG. **3A**, in response to the ratio being in the predetermined range, it may indicate that image **430** obtained by second object lens **410** has captured most target cells in subpart **353**.

(39) In some embodiments, in conjunction with FIG. **3A**, in response to the ratio not being in the predetermined range, it may indicate that image **430** obtained by second object lens **410** has not captured most target cells in subpart **353**. Therefore, second object lens **410** is further configured to focus on another secondary subpart (e.g., secondary subpart **423**) adjacent to secondary subpart **422** and obtain and save an image associated with secondary subpart **423**.

(40) FIG. **4C** illustrates how an example multiple object lens module obtains images associated with a plurality of secondary subparts (e.g., secondary subparts **422** and **423**) of a cytology specimen by a second object lens and FIG. **4D** illustrates an image **450** associated with secondary subpart **423** obtained by second object lens **410**, arranged in accordance with some embodiments of the present disclosure. Image **450** includes an image of target cell **451**. Image of target cell **451** may include the first specific range in RGB domain or in HSV domain.

(41) In some embodiments, in conjunction with FIGS. **2** and **3A**, images **430** and **450** obtained by second object lens **410** of the multiple object lens module are compared to image **333** obtained by first object lens **220** to determine whether a ratio of a number of target cells in images **430** and **450** to a number of target cells in image **333** is in a predetermined range. In conjunction with FIG. **3A**, in response to the ratio being in the predetermined range, it may indicate that images **430** and **450** obtained by second object lens **410** have captured most target cells in subpart **353**.

(42) In some embodiments, in conjunction with FIGS. **2** and **3A**, images **430** and **450** obtained by second object lens **420** of the multiple object lens module are compared to image **333** obtained by first object lens **220** to determine whether a ratio of a number of target cells in images **430** and **450** to a number of target cells in image **333** is in a predetermined range. In conjunction with FIG. **3A**, in response to the ratio being in the predetermined range, it may indicate that images **430** and **450** obtained by second object lens **410** have captured most target cells in subpart **353**.

(43) In response to the ratio not being in the predetermined range, it may indicate that images **430** and **450** obtained by second object lens **410** have not captured most target cells in subpart **353**. Then second object lens **410** is configured to focus on another secondary subpart (e.g., secondary subpart **421**) adjacent to subpart **422** and obtain and save an image associated with secondary subpart **421**. The process above may be repeated until a ratio of a number of target cells in images obtained by second object lens **410** to a number of target cells in image **333** is in the predetermined range.

(44) In some embodiments, similarly, in conjunction with FIG. 3A, second object lens **410** is configured to obtain images associated with other subparts (e.g., subparts **351** and **355** in FIG. 3A). As set forth above, second object lens **410** may be configured to obtain less images associated with other subparts (e.g., subparts **351** and **355** in FIG. 3A) than images associated with subpart **353/420** including the target cells. For example, compared to images **430** and **450** obtained by second object lens **410**, second object lens **410** is configured to obtain an image of one single secondary subpart of subpart **351** and an image of one single secondary subpart of subpart **355**.

(45) FIG. 5 illustrates an example system **500** to obtain images associated with a cytology specimen, arranged in accordance with some embodiments of the present disclosure. System **500** includes, but not limited to, computing device **510**, camera **520**, multiple object lens module **530**, stage **550** and light source **560**. In some embodiments, stage **550** is configured to carry and move a glass slide. Cytology specimen **540** is distributed on the glass slide.

(46) In some embodiments, computing device **510** includes a processor, a memory subsystem, and a communication subsystem. Artificial intelligence engines discussed above may be implemented as a set of executable instructions stored in the memory subsystem to be executed by the processor.

(47) In some embodiments, computing device **510** is configured to generate control signals and transmit the control signals to camera **520**, multiple object lens module **530**, stage **550** and light source **560** via the communication subsystem. For example, computing device **510** is configured to control light source **560** to generate lights with a specific range of wavelengths associated with image characteristics of target cells distributed in cytology specimen **540**. For example, the specific range of wavelengths may correspond to the color information associated with the target cells as set forth above. An example specific range of wavelengths may be about 530 nm to about 630 nm. Alternatively, another example specific range of wavelengths may be about 450 nm to about 560 nm, and preferably about 450 nm to about 530 nm.

(48) In some embodiments, computing device **510** is configured to control a movement of stage **550**. When stage **550** carries the glass slide and cytology specimen **540**, the movement of stage **550** may align a region (e.g., region A, B, C, D, E, F, G, H, I or J illustrated in FIG. 2 or subregion **341**, **343** or **345** in FIG. 3A) of cytology specimen **540** and a field of view of object lens **533** or **535** with a light path of the light generated by light source **560** and illustrated in FIG. 5. Therefore, the light generated by light source **560** may pass through the region and object lens **533** or **535** and reach camera **520**.

(49) In some embodiments, computing device **510** is configured to control lens switching module **531** of multiple object lens module **530** to switch between different object lens (e.g., object lens **533** and **535**). In some other embodiments, computing device **510** is configured to control camera **520**, multiple object lens module **530** and/or stage **550** to move so that object lens **533** or object lens **535** to focus on parts, subparts and/or secondary subparts of cytology specimen **540**. In conjunction with FIG. 2, object lens **533** may correspond to first object lens **220**. In conjunction with FIG. 4A, object lens **555** may correspond to second object lens **410**.

(50) In some embodiments, computing device **510** is configured to control camera **520** to obtain an image of a focused part of cytology specimen **540**. Computing device **510** is also configured to control camera **520** to send the obtained image to computing device **510** for further processing. Such processing includes, but not limited to, identifying images associated with target cells or non-target cells in cytology specimen **540**, comparing one or more images obtained by second object

lens **535** to an image obtained by first object lens **533** to determine whether a ratio of a number of target cells in the images obtained by second object lens **535** to a number of target cells in the image obtained by first object lens **533** is in a predetermined range.

(51) FIG. **6** is a flow diagram illustrating an example process **600** to obtain images associated with target cells distributed in a cytology specimen, arranged in accordance with some embodiments of the present disclosure. Process **600** may include one or more operations, functions, or actions as illustrated by blocks **610**, **620**, **630**, **640**, **650** and/or **660** which may be performed by hardware, software and/or firmware. The various blocks are not intended to be limiting to the described embodiments. The outlined steps and operations are only provided as examples, and some of the steps and operations may be optional, combined into fewer steps and operations, or expanded into additional steps and operations without detracting from the essence of the disclosed embodiments. Although the blocks are illustrated in a sequential order, these blocks may also be performed in parallel, and/or in a different order than those described herein.

(52) Process **600** may begin at block **610**, “obtain first image associated with first region of cytology specimen.” In some embodiments, in conjunction with FIG. **2** and FIG. **5**, processor **510** is configured to control a movement of stage **550** so that a first region (e.g., region H in FIG. **2**) of cytology specimen **210** is aligned with a first field of view of first object lens **220/533**. Processor **510** is configured to control light source **560** to generate lights with a specific range of wavelengths corresponding to the color information associated with target cells as set forth above in cytology specimen **210**. An example specific range of wavelengths may be about 530 nm to about 630 nm. Alternatively, another example specific range of wavelengths may be about 450 nm to about 560 nm, and preferably about 450 nm to about 530 nm. The lights may pass through the first region and first object lens **220/533**. Processor **510** is configured to control camera **520** to obtain images associated with the first region through first object lens **220/533**.

(53) In some embodiments, first object lens **220/533** has a first depth of field in the first field of view. The first depth of field may be less than a thickness of cytology specimen **210**. Therefore, to obtain sharp and focused images associated with the entire first region, camera **520** is configured to sequentially obtain images of different layers in the first region of cytology specimen **210** according to the first depth of field. For example, in conjunction with FIG. **2**, camera **520** is configured to sequentially obtain images of parts H1, H2 and H3. In some embodiments, processor **510** is configured to identify target cells and/or non-target cells from images of parts H1, H2 and H3 based on contrasts, lightness, shapes and other image characteristics associated with the target cells in cytology specimen **210**.

(54) Block **610** may be repeated to perform on different regions (e.g., regions F, I and J in FIG. **2**) of the cytology specimen. For example, processor **510** is configured to additionally obtain images of parts F1, F2, F3, I1, I2, I3, J1, J2 and J3 of cytology specimen **210** and identify target cells and/or non-target cells from images of F1, F2, F3, I1, I2, I3, J1, J2 and J3 based on contrasts, lightness, shapes and other image characteristics associated with the target cells and/or non-target cells in cytology specimen **210**.

(55) In some embodiments, in response to target cells are identified to be mostly dispersed in a specific layer (e.g., second layer) in the regions (e.g., regions F, H, I and J) of cytology specimen **210**, processor **510** is configured to obtain a first image (e.g., image of part H2) associated with respective region (e.g., region H) of cytology specimen. Eventually, for example, processor **510** is configured to obtain images of A2, B2, C2, D2, E2, F2, G2, H2, I2 and J2 associated with regions A, B, C, D, E, F, G, H, I and J, respectively.

(56) Block **610** may be followed by block **620**, “determine first subregion of first region.” In some embodiments, processor **510** is configured to sequentially identify images associated with the target cells and/or the non-target cells from images of A2, B2, C2, D2, E2, F2, G2, H2, I2 or J2. For example, in conjunction with FIG. **3A** and FIG. **3B**, processor **510** is configured to identify images **331**, **333** and **335**. Images associated with cells **321** and **322** are identified to be included in image

331, images associated with cells **323**, **324** and **326** are identified to be included in image **333** and images associated with cells **321**, **322** and **325** are identified to be included in image **335**. As previously discussed, based on spatial relationships between images **331**, **333**, **335** and image region H', and a correspondence between image region H' and region H, subregions **341**, **343** and **345** in region H may be identified. In conjunction with FIG. 2, block **620** may be repeated to perform on other images (e.g., images of **A2**, **B2**, **C2**, **D2**, **E2**, **F2**, **G2**, **12** and **J2**) associated with other regions (e.g., regions A, B, C, D, E, F, G, I and J) of cytology specimen **210**. Therefore, subregions of region A, B, C, D, E, F, G, I and J associated with the target cells and/or the non-target cells may also be determined in block **620**.

(57) Block **620** may be followed by block **630**, "obtain second set of images associated with first subregion of cytology specimen." In some embodiments, in conjunction with FIG. 3A, as previously discussed, a second object lens is configured to obtain more detailed images of cells **321** to **326** in first subregions **341**, **343** and **345**. In some embodiments, in conjunction with FIG. 4A and FIG. 5, processor **510** is configured to control a movement of stage **550** so that a first subregion (e.g., subregion **343** in FIG. 4A) of the cytology specimen is aligned with a second field of view of second object lens **410/535**. Processor **510** is configured to control light source **560** to generate lights with a specific range of wavelengths corresponding to the color information associated with target cells as set forth above in the cytology specimen. An example specific range of wavelengths may be about 530 nm to about 630 nm. Alternatively, another example specific range of wavelengths may be about 450 nm to about 560 nm, and preferably about 450 nm to about 530 nm. The lights may pass through the first subregion and second object lens **410/535**. Processor **510** is configured to control camera **520** to obtain a second set of images associated with the first subregion through second object lens **410/535**. The second set of images may include one or more images of secondary subparts **421**, **422**, **423** and **424** of subpart **420**.

(58) In some embodiments, in conjunction with FIG. 4A and FIG. 5, processor **510** is configured to control second object lens **410/535** to focus and obtain the second set of images based on image characteristics of target cells, for example, color information associated with the target cells. For example, the second set of images (e.g., image of secondary subpart **422**, meaning image **430**) includes the color information associated with the target cells but images associated with first subregion **343** other than the second set of images (e.g., images of secondary subparts **421**, **423** and **424**) do not include the color information associated with the target cells.

(59) Block **630** may be followed by block **640** "identify third image from first image." In some embodiments, in conjunction with FIG. 3A and 5, processor **510** is configured to identify third image **333** from image **330** based on contrasts, lightness, shapes and other image characteristics associated with cells **321** to **326**. Noting that third image **333** is a subset of first image **330** obtained through first object lens **220/533** and includes images of target cells distributed in subpart **353** defined by distance 2 and subregion **343**.

(60) Block **640** may be followed by block **650** "determine whether ratio is in predetermined range." In some embodiments, in conjunction with FIG. 3B and FIG. 4B, the ratio may be a ratio of a number of target cells in the second set of images discussed above (e.g., one or more images of secondary subparts **421**, **422**, **423** and **424**, image **430**) to a number of target cells in the third image (e.g., image **333**). In some embodiments, as discussed above, in conjunction with FIG. 3A and 4A, the second set of images (e.g., image of secondary subpart **422**, image **430**) is an image of subset of subpart **420/353**. Therefore, the third image (e.g., image **333**) of subpart **353** should include images of target cells not less than the second set of images (e.g., image **430**).

(61) In response to the ratio in the predetermined range, block **650** may be followed by block **660** "obtain image associated with target cells based on second set of images." In response to the ratio being in the predetermined range, it may indicate that the second set of images (e.g., image **430**) obtained by second object lens **410** include images of most target cells in subpart **353**. Therefore, images associated with target cells may be obtained based on the second set of images. In some

embodiments, for illustration only, the predetermined range is between around 50% and around 200%.

(62) In response to the ratio not being in the predetermined range, block **640** may return to block **630**. In conjunction with FIG. 3A, FIG. 4A and FIG. 4B, it may indicate that the second set of images (e.g., image **430**) obtained by second object lens **410** do not include images of most target cells in subpart **353**. Block **650** may be followed by block **630** to obtain additional image(s) associated with first subregion to be included in the second set of images. For example, in conjunction with FIG. 4A and FIG. 5, processor **510** is configured to control second object lens **410** to focus on another secondary subpart (e.g., subpart **423**) adjacent to secondary subpart **422**, obtain an image associated with secondary subpart **423** (i.e., image **450**) and include image **450** in the second set of images.

(63) The above examples can be implemented by hardware (including hardware logic circuitry), software or firmware or a combination thereof. The above examples may be implemented by any suitable computing device, computer system, wearables, etc. The computing device may include processor(s), memory unit(s) and physical NIC(s) that may communicate with each other via a communication bus, etc. The computing device may include a non-transitory computer-readable medium having stored thereon instructions or program code that, in response to execution by the processor, cause the processor to perform processes described herein with reference to FIGS. 2, 3, 5, and 6. For example, a computing device may communicate with a wearable and/or one or more sensors.

(64) The techniques introduced above can be implemented in special-purpose hardwired circuitry, in software and/or firmware in conjunction with programmable circuitry, or in a combination thereof. Special-purpose hardwired circuitry may be in the form of, for example, one or more application-specific integrated circuits (ASICs), programmable logic devices (PLDs), field-programmable gate arrays (FPGAs), and others. The term ‘processor’ is to be interpreted broadly to include a processing unit, ASIC, logic unit, or programmable gate array etc.

(65) Some aspects of the embodiments disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computing systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware are possible in light of this disclosure.

(66) Software and/or other instructions to implement the techniques introduced here may be stored on a non-transitory computer-readable storage medium and may be executed by one or more general-purpose or special-purpose programmable microprocessors. A “computer-readable storage medium”, as the term is used herein, includes any mechanism that provides (i.e., stores and/or transmits) information in a form accessible by a machine (e.g., a computer, network device, personal digital assistant (PDA), mobile device, manufacturing tool, any device with a set of one or more processors, etc.). A computer-readable storage medium may include recordable/non recordable media (e.g., read-only memory (ROM), random access memory (RAM), magnetic disk or optical storage media, flash memory devices, etc.)

(67) From the foregoing, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various embodiments disclosed herein are not intended to be limiting.

Claims

1. A method to obtain images associated with target cells distributed in a three-dimensional manner in a cytology specimen, comprising: obtaining a first image associated with a first region of the cytology specimen through a first object lens having a first field of view and a first depth of field in the first field of view, wherein a first layer of the cytology specimen corresponds to the first depth of field; determining a first subregion of the first region based on image characteristics associated with the target cells, wherein a first number of the target cells are distributed in a first space defined by the first subregion and the first depth of field, wherein the first space is associated with the first layer; obtaining a second set of one or more images associated with the first subregion through a second object lens having a second field of view and a second depth of field in the second field of view, wherein the second depth of field corresponds to a sublayer of the first layer; identifying a third image from the first image based on image characteristics associated with the target cells, wherein the third image is associated with the first subregion and covers the first number of the target cells; and obtaining images associated with the target cells based on the second set of one or more images in response to a comparison result between the second set of one or more images and the third image.
2. The method of claim 1, wherein determining the first subregion further comprises identifying the target cells from the first image based on the image characteristics.
3. The method of claim 1, wherein the image characteristics associated with the target cells include contrast, lightness, shape, and color information associated with the target cells.
4. The method of claim 3, wherein the color information corresponds to lights with a first range of wavelengths from about 530 nm to about 630 nm, a second range of wavelengths from about 450 nm to about 560 nm or a third range of wavelengths from about 450 nm to 530 nm.
5. The method of claim 3, wherein the color information includes a range of Red (R), Green (G), and Blue (B) values in an RGB domain or a range of Hue values in an HSV (Hue, Saturation, Value) domain.
6. The method of claim 1, wherein the second depth of field is less than the first depth of field.
7. The method of claim 1, wherein the second set of one or more images correspond to images obtained from different sublayers in the field of view.
8. The method of claim 1, wherein the third image is a subset of the first image.
9. The method of claim 1, wherein the comparison is associated with a ratio of a second number of the target cells in the second set of one or more images to the first number of the target cells in the third image being in a predetermined range.
10. The method of claim 1, wherein the second field of view corresponds to one or more parts of the first subregion.
11. The method of claim 1, prior to obtaining the first image, further comprising: obtaining images associated with other regions of the cytology specimen through the first object lens; determining the first layer of the cytology specimen that includes the target cells based on the images associated with the other regions; and obtaining images of the first layer through the first object lens.
12. The method of claim 1, wherein the obtaining the second set of one or more images further comprises: driving the second object lens to focus on a second space of the cytology specimen defined by the first subregion and the second depth of field; and obtaining an image of the second space to be one of the second set of one or more images, wherein the second space is associated with the sublayer.
13. The method of claim 1, further comprising, in response to the obtaining images associated with the target cells in an image associated with the first space, driving the second object lens to focus on a third space of the cytology specimen adjacent to the first space and obtaining an image of the third space to be one of the second set of one or more images.
14. A system to obtain images associated with target cells distributed in a three-dimensional manner in a cytology specimen, comprising: a processor; and a non-transitory computer-readable

medium having stored thereon instructions that, in response to execution by the processor, cause the processor to: obtain a first image associated with a first region of the cytology specimen through a first object lens having a first field of view and a first depth of field in the first field of view, wherein a first layer of the cytology specimen corresponds to the first depth of field; determine a first subregion of the first region based on image characteristics associated with the target cells, wherein a first number of the target cells are distributed in a first space defined by the first subregion and the first depth of field, wherein the first space is associated with the first layer; obtain a second set of one or more images associated with the first subregion through a second object lens having a second field of view and a second depth of field in the second field of view, wherein the second depth of field corresponds to a sublayer of the first layer; identify a third image from the first image based on image characteristics associated with the target cells, wherein the third image is associated with the first subregion and covers the first number of the target cells; and obtain images associated with the target cells based on the second set of one or more images in response to a comparison between the second set of one or more images and the third image.

15. The system of claim 14, wherein the non-transitory computer-readable medium having stored thereon additional instructions that, in response to execution by the processor, cause the processor to identify the target cells from the first image based on the image characteristics.

16. The system of claim 14, wherein the second set of one or more images correspond to images obtained from different sublayers in the second field of view.

17. The system of claim 14, wherein the third image is a subset of the first image.

18. The system of claim 14, wherein the comparison is associated with a ratio of a second number of the target cells in the second set of one or more images to the first number of the target cells in the third image being in a predetermined range.

19. The system of claim 14, wherein the non-transitory computer-readable medium having stored thereon additional instructions that, in response to execution by the processor, cause the processor to: drive the second object lens to focus on a second space of the cytology specimen defined by the first subregion and the second depth of field; and obtain an image of the second space to be one of the second set of one or more images, wherein the second space is associated with the sublayer.

20. The system of claim 14, wherein the non-transitory computer-readable medium having stored thereon additional instructions that, in response to execution by the processor, cause the processor to: in response to the obtaining images associated with the target cells in an image associated with the first space, drive the second object lens to focus on a third space of the cytology specimen adjacent to the first space and obtain an image of the third space to be one of the second set of one or more images.
